

2024 AMC 10A Problem 12 - Solution

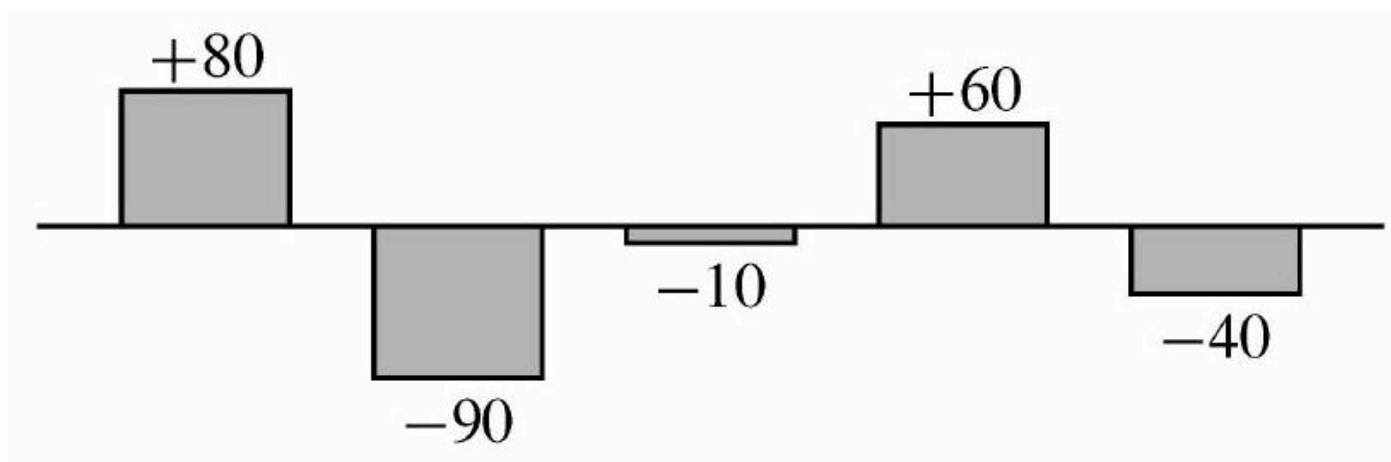
Review the full statement and step-by-step solution for 2024 AMC 10A Problem 12. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10A Problem 12

Problem:

Zelda played the Adventures of Math game on August 1 and scored 1700 points. She continued to play daily over the next 5 days. The bar chart below shows the daily change in her score compared to the day before. (For example, Zelda's score on August 2 was $1700 + 80 = 1780$ points.) What was Zelda's average score in points over the 6 days?

Daily Change in Score from August 2 to 6



Aug 2 Aug 3 Aug 4 Aug 5 Aug 6

Answer Choices:

A. 1700

B. 1702

C. 1703

D. 1713

E. 1715

Solution:

The table below shows Zelda's daily scores over the 6 days.

Date	Change	Score
<i>Aug1</i>		1700
<i>Aug2</i>	+80	1780
<i>Aug3</i>	−90	1690
<i>Aug4</i>	−10	1680
<i>Aug5</i>	+60	1740
<i>Aug6</i>	−40	1700

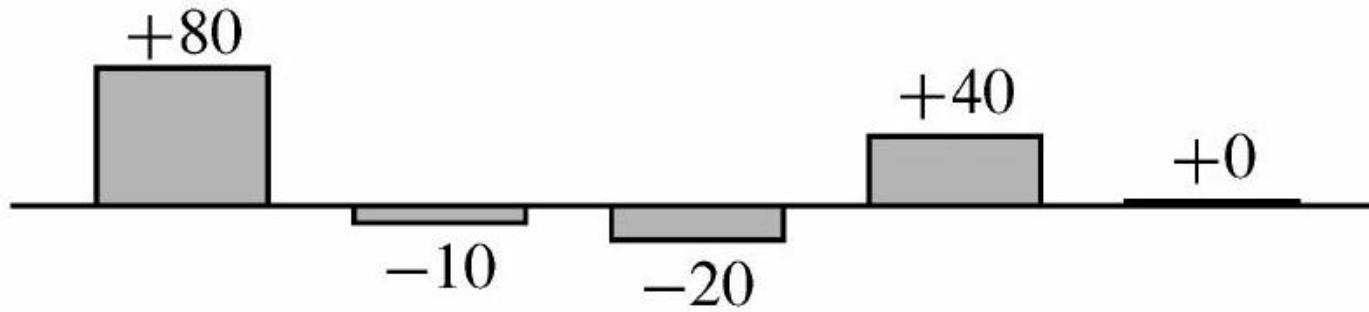
Summing the scores and dividing by 6 gives an average of

$$\frac{1700 + 1780 + 1690 + 1680 + 1740 + 1700}{6} = \frac{10290}{6} = \text{(E)} \boxed{1715} \text{ points.}$$

OR

The bar chart below shows the cumulative change in Zelda's score relative to the 1700 points scored on August 1.

Cumulative Change in Score since August 1



Aug 2 Aug 3 Aug 4 Aug 5 Aug 6

The average cumulative change over the 6 days was

$$\frac{80 - 10 - 20 + 40 + 0}{6} = \frac{90}{6} = 15 \text{ points}$$

so Zelda's average score over the 6 days was $1700 + 15 = (E) \boxed{1715}$ points.

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